

Quantum Three-Body Reactions in Nuclear Physics: Five Hallmarks of a Unified Reaction Theory

Jin Lei¹

¹*Department of Physics,
Tongji University,
Shanghai 200092,
China*

(Dated: June 9, 2026)

The quantum three-body problem sits at the center of low-energy nuclear reaction theory. Whenever a weakly bound projectile such as the deuteron, ${}^6\text{Li}$, ${}^{11}\text{Be}$, or ${}^6\text{He}$ strikes a target, the relevant dynamics is irreducibly three-body: a core, a valence cluster, and the target, coupled through breakup, transfer, and fusion. A vast and seemingly heterogeneous body of methods has grown to describe these reactions, from the exact Faddeev and Alt-Grassberger-Sandhas equations through the continuum-discretized coupled-channels framework to closed-form inclusive-breakup sum rules, semiclassical and eikonal models, and, most recently, reduced-basis emulators and differentiable Bayesian calibration. This review argues that the apparent diversity collapses onto a small number of organizing principles. I identify five hallmarks that a complete three-body reaction theory must possess: exactness against the Faddeev/AGS standard; discretization of the breakup continuum; dynamic coupling to that continuum; the treatment of inclusive and spectator processes; and the universality that connects nuclear three-body observables to a wider few-body physics. Two enabling characteristics, the control of three-body asymptotics and boundary conditions, and the modern machinery of computation, emulation, and inference, cut across all five. Each hallmark is developed as a self-contained essay with its mathematical formulation, methodological landscape, landmark results, and open questions, and the hallmarks are then woven into a single picture in which the same breakup amplitude, viewed through different observables, produces elastic scattering, breakup spectra, fusion suppression, and inclusive cross sections. I close with the open challenges, principally the four-body frontier, the consistent merger of exact dynamics with realistic structure, and the calibration of expensive three-body models against data with honest uncertainties, that will define the next decade.

CONTENTS

		C. Why universality matters for reactions	8
I. Introduction	1	VII. Enabling characteristic 1: Asymptotics and boundary conditions	8
II. Hallmark 1: Exactness and the Faddeev/AGS standard	3	VIII. Enabling characteristic 2: Computation, emulation, and inference	9
A. Why a three-body Lippmann-Schwinger equation is not enough	3	A. The cost problem and the reduced-basis answer	9
B. The momentum-space program and its benchmarks	3	B. Scattering emulators and the differentiable turn	9
C. The Coulomb problem	3	C. Inference and its pitfalls	10
D. Universality through the exact lens, and the cost barrier	3	IX. Synthesis: one amplitude, many observables	10
III. Hallmark 2: Discretizing the continuum with CDCC	4	X. Open challenges	10
A. The construction	4	XI. Conclusions	11
B. What CDCC computes, and what it reduces to	5	Acknowledgments	11
C. The CDCC family tree	5	References	11
IV. Hallmark 3: Dynamic coupling to breakup	5		
A. The polarizable continuum	5		
B. Fusion suppression and threshold anomalies	6		
C. Mechanism, not just magnitude	6		
V. Hallmark 4: Inclusive and spectator processes	7		
A. Summing over the unobserved fragment	7		
B. The post-prior controversy and its resolution	7		
C. From DWBA entrance to CDCC entrance	7		
D. Spectroscopy, knockout, and surrogates	8		
VI. Hallmark 5: Universality and scale separation	8		
A. The organizing idea	8		
B. Effective field theory and the nuclear cases	8		

I. INTRODUCTION

The motion of three mutually interacting bodies has resisted closed-form solution since Newton, and its quantum version is no less unforgiving. What is remarkable is how often the three-body problem is the *right* problem in nuclear physics. The deuteron is a proton and a

neutron so loosely bound that any collision with a nucleus is, from the first femtosecond, a problem of three bodies in relative motion. The lithium isotopes ${}^6\text{Li}$ and ${}^7\text{Li}$ are an α particle plus a deuteron or a triton, held together by less than two MeV. The one-neutron halo ${}^{11}\text{Be}$ is a ${}^{10}\text{Be}$ core orbited by a single neutron whose wave function extends far beyond the nuclear surface; the Borromean halos ${}^6\text{He}$ and ${}^{11}\text{Li}$ fall apart if any one of their three constituents is removed (Zhukov *et al.*, 1993). For all of these, the reaction with a target nucleus cannot be reduced to two-body elastic scattering plus perturbative corrections. The breakup channel is open at essentially all energies, it is strongly coupled, and it reshapes every other observable.

This recognition is old. The earliest serious three-body treatments of deuteron stripping date to the 1960s and 1970s, and by the late 1980s the field possessed both an exact formalism, the integral equations of Faddeev (Faddeev, 1961) and their reorganization by Alt, Grassberger, and Sandhas (Alt *et al.*, 1967), and a practical approximation, the continuum-discretized coupled-channels (CDCC) method codified in the review of Austern and collaborators (Austern *et al.*, 1987). In the decades since, the advent of radioactive-ion beams has turned three-body reaction theory from a specialist topic into a central tool: the structure of nuclei far from stability is read off largely through reactions, and almost every exotic projectile of interest is weakly bound. The result is a literature of striking breadth. Exact few-body practitioners solve momentum-space integral equations with separable interactions (Glöckle *et al.*, 1996); reaction phenomenologists discretize the continuum into bins and solve coupled differential equations (Thompson, 1988); nuclear astrophysicists invert breakup measurements to extract capture cross sections (Typel and Baur, 1994; Tribble *et al.*, 2014); and a new generation builds fast surrogates of these calculations to enable statistical inference (Driscler *et al.*, 2023).

Confronted with this landscape, a newcomer could be forgiven for seeing a collection of unrelated techniques, each with its own jargon, its own benchmark systems, and its own community. The thesis of this review is the opposite. Beneath the methodological variety lies a small set of physical and computational principles, each of which is a capability the field has had to acquire in order to turn a formal three-body scattering problem into a quantitative predictor of laboratory observables. These principles are not themselves methods; they are the questions every method must answer. How do we know an approximate calculation is correct? How do we make a continuum tractable? How does the continuum react back on the channels we measure? How do we connect the amplitude to the inclusive cross section that the detector actually records? And what, if anything, is universal across the bewildering variety of weakly bound systems?

I organize the answers into *five hallmarks*. The first is

exactness: the Faddeev/AGS equations are the standard of correctness, the unique formulation in which all rearrangement channels and their asymptotics are treated on an equal footing and the disconnected-kernel pathology of a naive three-body Lippmann-Schwinger equation is removed. They are expensive, and for many systems impractical, but they are what every approximation is measured against. The second is the **discretization of the continuum:** CDCC replaces the non-denumerable breakup continuum by a finite, square-integrable basis, converting a three-body scattering problem into a finite set of coupled-channels equations that standard reaction machinery can solve. The third is **dynamic coupling to breakup:** the central physics of weakly bound systems is that the continuum is not a passive spectator but a polarizable medium that back-reacts on elastic scattering, transfer, and fusion through a dynamic polarization potential, producing fusion suppression, threshold anomalies, and characteristic breakup-energy distributions. The fourth is **inclusive and spectator processes:** summing coherently over an unobserved fragment yields non-elastic breakup, incomplete fusion, knockout, and the closed-form inclusive sum rules that bridge three-body amplitudes to the inclusive cross sections actually measured. The fifth is **universality and scale separation:** near threshold, three-body observables are organized by a few large scales, the scattering lengths and the Efimov geometric tower, nearly independent of short-range detail, and this connects nuclear three-body reactions to cold atoms, hypernuclei, and the wider physics of few-body universality.

Two further themes are not hallmarks because they do not sit at the same level of abstraction; they are *enabling characteristics* that cut across all five. The first is the control of **asymptotics and boundary conditions:** three-body Coulomb has no closed-form asymptotic form, the breakup continuum makes outgoing-wave matching subtle, and every hallmark therefore rests on a specific boundary-condition technology, complex scaling, the Lagrange-mesh R-matrix, screening and renormalization, or the Vincent-Fortune contour. The second is **computation, emulation, and inference:** a single high-fidelity three-body solve is expensive, calibration and uncertainty quantification require many thousands of solves, and the reduced-basis emulators, differentiable solvers, and gradient-based samplers that have appeared in the past five years are what convert a forward map into a calibrated prediction with error bars.

The remainder of the review develops each hallmark in turn (Secs. II–VI), then the two enabling characteristics (Secs. VII–VIII). Section IX draws the connections: I argue that the same breakup amplitude, projected onto different final-state conditions, generates the elastic, breakup, fusion, and inclusive observables that the five hallmarks treat separately, and I give practical guidance on which method answers which question. Sec-

tion X states the open challenges as a sequence of focused problems, and Sec. XI offers a forward-looking view of where the field is heading. Figure 1 is the visual thesis: the five hallmarks arranged around the central object, the three-body reaction observable, resting on the two enabling characteristics that make them computable.

II. HALLMARK 1: EXACTNESS AND THE FADDEEV/AGS STANDARD

A. Why a three-body Lippmann-Schwinger equation is not enough

The natural first attempt at three-body scattering, writing a Lippmann-Schwinger equation for the full three-body transition operator, fails for a reason that is structural rather than numerical. The kernel of that equation contains disconnected pieces in which one of the three particles is a pure spectator, and these pieces prevent the kernel from being compact; the equation has no unique solution. Faddeev’s insight (Faddeev, 1961) was to decompose the wave function into three components labeled by which pair is interacting, and to write coupled integral equations for those components whose kernel, after one iteration, is connected. Alt, Grassberger, and Sandhas (Alt *et al.*, 1967) recast the Faddeev equations as multichannel two-body Lippmann-Schwinger equations for transition operators between two-cluster partitions, a form especially convenient for nuclear applications, and Lovelace gave an early practical theory of three-particle states (Lovelace, 1964). The textbook synthesis of this machinery is Glöckle’s monograph (Glöckle, 1983).

B. The momentum-space program and its benchmarks

For nuclear systems the dominant production route solves the AGS equations in momentum space, almost always representing the two-body t -matrices by separable expansions of Ernst-Shakin-Thaler type (Ernst *et al.*, 1973) so that the integral equations reduce to manageable linear systems. The reference statement of this program for the three-nucleon continuum is the Bochum review of Glöckle and collaborators (Glöckle *et al.*, 1996), which fixes the accuracy standard for nucleon-deuteron elastic scattering and breakup with realistic two-nucleon forces and the two-pion-exchange three-nucleon force, and which frames the still-open puzzles of the field: the roughly thirty-percent underprediction of the low-energy vector analyzing power that current three-nucleon forces do not cure, the space-star and quasi-free breakup anomalies, and the Phillips-line correlation between the doublet nucleon-deuteron scattering length and the triton binding energy.

The role of these exact solutions in the broader reaction-theory enterprise is as a *benchmark*. The canon-

ical head-to-head comparisons of CDCC against the exact Faddeev/AGS solution for deuteron-induced reactions (Deltuva *et al.*, 2007; Upadhyay *et al.*, 2012) are what license the use of CDCC across the much larger space of systems where an exact calculation is out of reach. When CDCC and AGS agree on elastic, transfer, and breakup observables for the same three-body Hamiltonian, the approximation is validated; where they disagree, the disagreement localizes the physics that the approximation omits. This benchmark relationship is the first of the cross-links in Fig. 1.

C. The Coulomb problem

The chief obstacle to exact three-body calculations for charged systems is that the long-range Coulomb interaction destroys the compactness of the Faddeev kernel: the asymptotic motion of a charged spectator pair is not free. Three resolutions coexist. The Lisbon group screens the Coulomb potential and renormalizes, which underlies the benchmark calculations above (Deltuva *et al.*, 2007). Because screening becomes intractable for high- Z targets, the TORUS collaboration developed a screening-free route in which the Coulomb interaction is carried in a distorted-wave basis (Mukhamedzhanov *et al.*, 2012). The Grenoble line instead follows Faddeev and Merkuriev, splitting each pair Coulomb potential into short- and long-range parts and absorbing the long-range tail into the asymptotic Coulomb wave (Merkuriev, 1980), which restores compact kernels and is solved in configuration space with complex scaling (Lazauskas and Carbonell, 2011).

D. Universality through the exact lens, and the cost barrier

Exact few-body methods are also the cleanest probe of three-body universality. Treating ${}^6\text{Li}$ as an effective neutron-proton-alpha three-body system and solving the separable Faddeev/AGS equations, the deuteron-alpha S-wave scattering length and the ${}^6\text{Li}$ three-body separation energy were shown to be remarkably insensitive to the choice of two-body interactions, leading to the characterization of ${}^6\text{Li}$ as a deuteron halo (Lei *et al.*, 2018). This result rests on a prior demonstration that the separable expansion reproduces direct nonseparable Faddeev solutions for the ${}^6\text{Li}$ bound state to four significant figures (Hlophe *et al.*, 2017) and for deuteron-alpha elastic and breakup observables at the one-percent level (Hlophe *et al.*, 2019), and it is the bridge from this hallmark to universality (Sec. VI). The leading chiral three-nucleon force enters this same continuum through the founding chiral analyses of nucleon-deuteron scattering (Epelbaum *et al.*, 2002).

The price of exactness is steep. A single Faddeev solu-

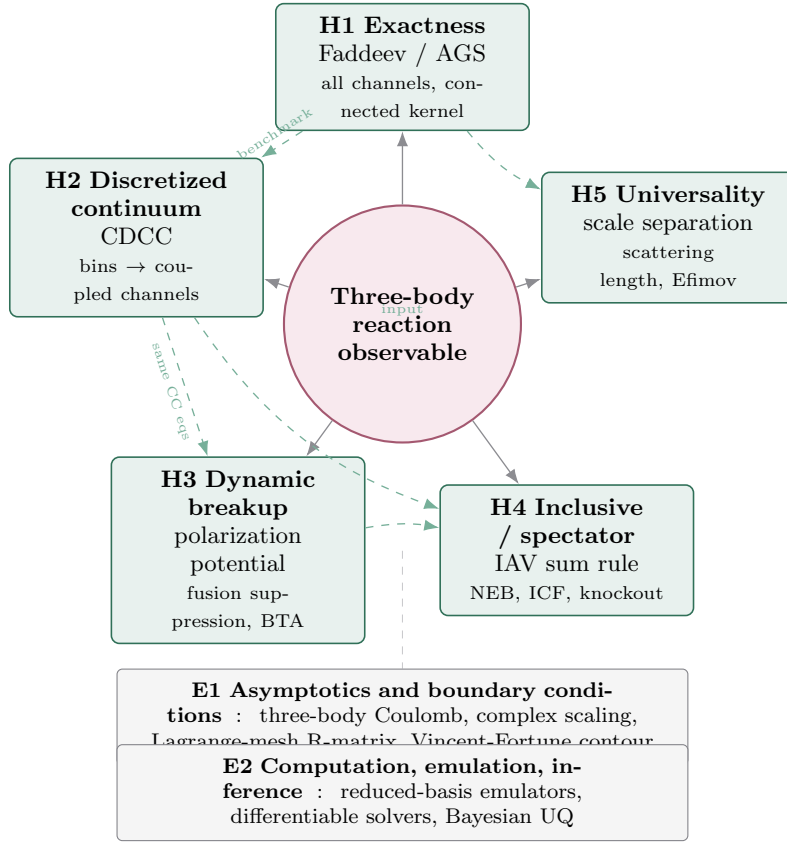


FIG. 1 The five hallmarks of quantum three-body reaction theory arranged around the central object they all serve, the three-body reaction observable. Green dashed links mark the principal cross-connections developed in Sec. IX: H1 benchmarks H2; H2 and H3 are the same coupled-channels equations read with and without the polarization interpretation; H2 and H3 feed the inclusive observables of H4 through the same breakup amplitude; and H5 supplies the universal low-energy input that constrains the interactions used in H1. The two enabling characteristics E1 and E2 form the substrate on which all five rest.

tion can cost on the order of 10^3 seconds per parameter point, and the four-body Yakubovsky generalization is dramatically harder, which is the principal motivation for the emulators discussed in Sec. VIII. Exactness thus sets the standard of correctness but cannot, by itself, span the space of systems and the space of parameters that modern applications demand. That tension, between the method that is right and the method that is affordable, runs through the entire review.

III. HALLMARK 2: DISCRETIZING THE CONTINUUM WITH CDCC

A. The construction

The continuum-discretized coupled-channels method makes the three-body scattering problem finite by two truncations (Austern *et al.*, 1987). The projectile internal continuum is truncated in angular momentum and discretized into a finite set of momentum bins on a bounded interval; each bin is a square-integrable wave packet, orthogonal to the bound states and to the other bins. For

a deuteron incident on a target, the three-body wave function is expanded on the deuteron ground state plus these bin states, and the effective Hamiltonian carries the proton-target and neutron-target optical potentials evaluated at half the deuteron energy. Projecting the three-body Schrödinger equation onto the truncated-and-discretized basis yields a finite set of coupled differential equations in the projectile-target coordinate, solvable with standard coupled-channels machinery. The continuum-bin construction, with its phase-shift or t -matrix weighting that is essential for representing narrow resonances, is the technical heart of the method (Thompson, 1988). The roots of the construction lie in the three-body deuteron-breakup-and-stripping analyses of the late 1970s (Austern *et al.*, 1978), and an alternative to the momentum-bin discretization, a Gaussian-expansion pseudostate basis that handles nuclear and Coulomb breakup on the same footing, has been developed in parallel (Egami *et al.*, 2004).

B. What CDCC computes, and what it reduces to

CDCC delivers elastic scattering with breakup back-coupling, the exclusive and energy-differential elastic-breakup cross sections, and, through one iteration of the post-form transition operator, the stripping amplitude. Several familiar methods are limits of CDCC: the adiabatic approximation is CDCC with all bin energies set equal; the single-channel projectile-folding optical model is what the elastic channel reduces to when continuum couplings are switched off; and the distorted-wave Born approximation is the linearization of the CDCC equations about the elastic channel. The Austern review established the structural facts that make the method work, in particular the suppression of long-range continuum-continuum coupling that keeps finite-radius outgoing-wave matching well defined, the necessity of the continuum-continuum couplings for the elastic channel, and the peripheral character of breakup with realistic absorptive potentials. The physical content of these couplings, in particular the way coupling to the continuum redistributes flux and modifies the elastic and inelastic channels, has been dissected in detail (Nunes *et al.*, 2004; N *et al.*, 2003), and the same CDCC wave function feeds the distorted-wave treatment of transfer, where core excitation can be included through an extended Born approximation (Gómez-Ramos *et al.*, 2015).

C. The CDCC family tree

The original formulation omitted Coulomb breakup and treated the target as inert; the subsequent half-century has been a steady relaxation of these restrictions. The extended CDCC line incorporates projectile core excitation and Coulomb breakup. Target collective excitation was brought into the coupled-channels set on an equal footing with projectile breakup, with the no-excitation limit reducing exactly to standard CDCC and validation against the exact Faddeev/AGS solution for deuteron scattering on ^{24}Mg (Gómez-Ramos and Moro, 2017). The four-body extension treats genuine three-body projectiles such as ^6He , ^6Li , and ^{11}Li , resolving the internal two-cluster subsystem explicitly (Matsumoto *et al.*, 2004; Rodríguez-Gallardo *et al.*, 2008); its canonical demonstration reproduces ^6Li elastic scattering on heavy targets with no adjustable parameter and reveals that of the many breakup pseudostates only the deuteron-alpha-dominant ones matter for elastic scattering (Watanabe *et al.*, 2015). Microscopic CDCC builds the projectile from an antisymmetrized cluster model and double-folds the nucleon-nucleon interaction, a development that resolved the long-standing renormalization anomaly of the folding model by tracing it to virtual breakup (Sakuragi *et al.*, 1986), and the importance of nonresonant continuum states beyond the prominent

resonances was established through polarized-beam observables (Hirabayashi, 1991; Kubono *et al.*, 1985). Relativistic CDCC extends the framework to the several-hundred-MeV-per-nucleon energies of radioactive-beam facilities (Aumann and Bertulani, 2020), and a Lagrange-mesh implementation has broadened the method's reach to exotic systems (Druet and Descouvemont, 2012). The convergence of the Weinberg-state expansion of the three-body wave function underpins the rapid convergence of these coupled-channels schemes for transfer (Pang *et al.*, 2013), and the modern review of the method and its applications is that of Yahiro and collaborators (Yahiro *et al.*, 2012).

The applications that have shaped the method are as instructive as the formalism. The breakup of the proton-rich ^8B has been a recurring testing ground, both as a three-body observable and as a probe of higher-order and E2 effects (Tostevin *et al.*, 2001; Mortimer *et al.*, 2002; Ogata *et al.*, 2006), and the breakup of the one-neutron halo ^{11}Be has driven the coupled-channels and time-dependent treatments of Coulomb-dominated dissociation (Howell *et al.*, 2005; Capel *et al.*, 2004). Extracting the resonance subspace from the full discretized continuum and comparing its effect against the nonresonant background has become a standard diagnostic (Carmacho *et al.*, 2015, 2016). On the experimental side, a large body of precision elastic-scattering and reaction-cross-section data on weakly bound and halo projectiles, the deuteron, $^6,7\text{Li}$, ^9Be , ^8B , ^6He , ^{10}Be , ^{17}F , and ^{11}Be on targets from carbon to lead, has both motivated and tested the CDCC family (R.C. *et al.*, 1997; Pietro *et al.*, 2004; Di Pietro *et al.*, 2010; Kolata *et al.*, 2004; Romoli *et al.*, 2004; Woolliscroft *et al.*, 2004; Benjamim *et al.*, 2007; Mohr *et al.*, 2010; Aguilera *et al.*, 2009; Zamora *et al.*, 2011; Barioni *et al.*, 2011).

The result is that CDCC is less a single method than a family, unified by the bin-discretization idea and differentiated by which degrees of freedom are resolved. Its accuracy where exact benchmarks exist, and its applicability where they do not, are what make it the workhorse of direct-reaction phenomenology.

IV. HALLMARK 3: DYNAMIC COUPLING TO BREAKUP

A. The polarizable continuum

The defining physics of a weakly bound projectile is that its breakup continuum is easy to excite and, once excited, reacts back on the channels one measures. This back-reaction is summarized by a dynamic polarization potential: a wave-function-equivalent local potential which, added to the bare optical potential, reproduces in a single-channel calculation the elastic wave function of the full coupled-channels solution (Thompson

et al., 1989). The founding observation, from microscopic CDCC, is that the polarization potential generated by coupling to the breakup continuum is qualitatively different from that generated by coupling to bound collective states: it is repulsive in its real part and only weakly absorptive, where collective coupling gives an attractive real and absorptive imaginary part (Sakuragi *et al.*, 1986). This single fact organizes much of the phenomenology of weakly bound systems.

B. Fusion suppression and threshold anomalies

The repulsive real polarization raises the effective Coulomb barrier and thereby suppresses complete fusion, an effect quantified in the survival-probability formalism in which the complete-fusion cross section acquires a partial-wave survival factor set by the strength of the coupling to breakup (Canto *et al.*, 1998; Hussein, 1984). The companion incomplete-fusion cross section factorizes transparently into a breakup probability, the fusion of one fragment, and the survival of the other (Yadav *et al.*, 2012). The experimental establishment of this suppression, against the broader backdrop of sub-barrier fusion enhancement by channel coupling (M, 1985; Balantekin and Takigawa, 1998; Rowley *et al.*, 1996; Gasques *et al.*, 2006), came from precision fusion measurements on heavy targets that disentangled complete from incomplete fusion: the landmark ${}^9\text{Be}+{}^{208}\text{Pb}$ result and the subsequent ${}^{6,7}\text{Li}$ and ${}^9\text{Be}$ systematics established a roughly thirty-percent complete-fusion suppression above the barrier (M *et al.*, 1999, 2002; Dasgupta *et al.*, 2004; Rath *et al.*, 2009), while the Borromean halo ${}^6\text{He}$ produced instead a sub-barrier enhancement (Trotta *et al.*, 2000). Coupled-channels calculations that include the discretized continuum reproduce and interpret these trends (Diaz-Torres and Thompson, 2002; Diaz-Torres *et al.*, 2003; K *et al.*, 2000; N *et al.*, 2001; Gomes *et al.*, 2005; Beck *et al.*, 2007; Beck, 2007; Beck *et al.*, 2010; Souza *et al.*, 2009; Lubian *et al.*, 2009; Canto *et al.*, 2009a; Aguilera *et al.*, 2000).

The same repulsive polarization inverts the usual near-barrier threshold anomaly of the optical potential into the breakup threshold anomaly characteristic of weakly bound projectiles, in which the expected rise of the real potential toward the barrier is cancelled while the imaginary part stays finite or rises. The threshold anomaly itself, the dispersion-relation link between the energy dependence of the real and imaginary optical potentials, was established for tightly bound heavy ions (Satchler *et al.*, 1987; Satchler, 1987, 1991; Keeley *et al.*, 1995), and its breakup-driven inversion was identified as a distinct manifestation of the same dispersion relation (Hussein *et al.*, 2006). A large systematic effort across light, medium, and heavy targets, ${}^{6,7}\text{Li}$ and ${}^9\text{Be}$ on aluminium, nickel, zinc, zirconium, tin, samarium, lead, and bis-

muth, has mapped where the anomaly is present, absent, or inverted (Gomes *et al.*, 2004; Figueira *et al.*, 2006, 2007; Kumawat *et al.*, 2008; Camacho *et al.*, 2010; Deshmukh *et al.*, 2011a,b; A *et al.*, 1998; Zadro *et al.*, 2009; Figueira *et al.*, 2010; Signorini *et al.*, 2002; Biswas *et al.*, 2008), with the closely related dipole polarizability and continuum-coupling effects probed through sub-barrier elastic scattering and the comparison of mirror-threshold systems (Parker *et al.*, 2008; Keeley *et al.*, 2002; Rusek *et al.*, 2003; Keeley *et al.*, 2003; Pakou *et al.*, 2003; Pakou and Rusek, 2004). Much of this work was carried out by the Brazilian-Mexican coupled-channels school, whose signature is the decomposition of the polarization potential and the extraction of the resonance subspace from the full continuum (Canto *et al.*, 2009b; Gómez Camacho *et al.*, 2018), and the field is reviewed comprehensively in two long articles (Canto *et al.*, 2006, 2015) and in the survey of fusion and direct reactions with halo nuclei (Keeley *et al.*, 2009).

C. Mechanism, not just magnitude

A recurring lesson of this hallmark is that a good reproduction of a cross section does not by itself identify the mechanism. The fusion suppression of ${}^{6,7}\text{Li}$ can be reproduced by a phenomenological repulsive polarization, but its microscopic origin is the loss of flux to the channel in which one fragment is captured and the other survives; framing this as a Trojan-Horse-like spectator effect ties the macroscopic flux-stealing picture to the microscopic inclusive-breakup mechanism of the next hallmark (Lei and Moro, 2019a), and resolves the long-standing puzzle of large alpha yields in lithium-induced reactions across the periodic table (Lei and Moro, 2017). The most recent development of this theme is a flux-conservation analysis of the coupled-channels absorption cross section through a generalized optical theorem, which decomposes the absorption into direct, breakup, and interference pieces and shows that neglecting the off-diagonal imaginary couplings biases the decomposition substantially (Liu *et al.*, 2026a,b). Whether the breakup that drives all of this is prompt, occurring at the point of closest approach, or delayed, following the decay of a populated resonance, is itself measurable through fragment-coincidence kinematics, and the answer bears directly on how the breakup couples to fusion (Luong *et al.*, 2011; So *et al.*, 2011).

The breakup-as-probe inverse of all this is Coulomb dissociation, in which the breakup of a projectile in the field of a high- Z target is inverted to extract a radiative-capture cross section for astrophysics. The early demonstration that breakup is peripheral and dominated by the projectile's surface (Baur *et al.*, 1980) grew into a quantitative method, most famously for the solar reaction ${}^7\text{Be}(p,\gamma){}^8\text{B}$ inferred from ${}^8\text{B}$ dissociation (Ogata *et al.*, 2006; Motobayashi *et al.*, 1994; Schumann *et al.*,

2003; Schümann *et al.*, 2006), with the careful accounting of higher-order, E2, and post-acceleration corrections that the three-body dynamics controls (Typel and Baur, 1994; Mortimer *et al.*, 2002; Esbensen and Bertsch, 1996; Kikuchi *et al.*, 1997; Davids *et al.*, 2001; Esbensen and Bertsch, 1999). The competition between nuclear and Coulomb breakup, and the multistep nature of the dissociation, must be modeled consistently for the inversion to be reliable (Capel *et al.*, 2004; Dasso *et al.*, 1998; Nunes and Thompson, 1999; Esbensen and Hencken, 2000; Kolata *et al.*, 2001; Descouvemont, 2004), and the method continues to deliver dipole-strength distributions for halo and Borromean systems at relativistic beam energies (Aumann and Bertulani, 2020; Lehr *et al.*, 2026).

V. HALLMARK 4: INCLUSIVE AND SPECTATOR PROCESSES

A. Summing over the unobserved fragment

Most measurements do not detect both outgoing fragments. When only one fragment is observed and one sums coherently over all final states of the other, the relevant object is the inclusive non-elastic breakup cross section, and its closed-form expression is the achievement of this hallmark. The Austern-Vincent sum rule (Austern and Vincent, 1981) and its definitive form due to Ichimura, Austern, and Vincent (Ichimura *et al.*, 1985) express the breakup-fusion cross section as the expectation value of the imaginary part of the unobserved fragment's optical potential, sandwiched between optical Green's functions and the source function that encodes the projectile breakup vertex. The construction replaces an explicit sum over breakup channels by a single Green's-function expectation, and it requires no explicit nuclear-structure input because the target internal coordinates are eliminated in favor of an empirical optical potential. Its conceptual ancestor is the inclusive-reaction theory of Kerman and McVoy (Kerman and McVoy, 1979), and the same physics was developed in parallel in the Jülich-Basel post-form distorted-wave program for deuteron and ^3He breakup (Baur *et al.*, 1983; Shyam *et al.*, 1980) and in the prior-form breakup-fusion description of nonequilibrium particle emission (Udagawa *et al.*, 1984). The first numerical test of the closed-form sum rule against an exact calculation, which also exposed the limits of the surface approximation, dates to the same period (Kasano and Ichimura, 1982).

B. The post-prior controversy and its resolution

For three decades two schools derived this sum rule in apparently incompatible forms, the post-form route of Austern, Vincent, Kasano, and Ichimura and the prior-form route of Udagawa and Tamura (Udagawa and

Tamura, 1981; Li *et al.*, 1984). The dispute turned on a nonorthogonality term and on an alleged ambiguity in the operator derivation, and the prior-form school exhibited cases in which the post-form approximations overshot the full calculation by two orders of magnitude. Ichimura, Austern, and Vincent resolved the matter analytically (Ichimura *et al.*, 1985): the corrected prior-form sum rule reduces exactly to the post-form one, the two schools' competing objects are the same mathematical quantity viewed through different interpretive lenses, and the convergence of the marginally convergent post-form integral is rigorously controlled by the Vincent-Fortune contour procedure. The two-step heuristic of breakup followed by absorption is valid in post form and invalid in prior form, an instructive reversal of the usual expectation that the physical picture is formalism-independent.

The analytic equivalence does not guarantee that practical implementations agree, since truncations and potential choices could leave residual discrepancies. The numerical resolution, that post and prior forms agree at the percent level wherever the analytic equivalence holds, was supplied three decades after the proof (Lei and Moro, 2015a, 2018), building on a transparent rederivation of the sum rule through the coupled-channels optical theorem (Lei and Moro, 2015b; Cotanch, 2010). An independent route to the same reconciliation, framed in the language of surrogate reactions and the nonorthogonality function, was given in parallel (Potel *et al.*, 2015). This is the kind of result that closes a chapter: a controversy that had generated a sizable literature was settled by computing both sides of an equality.

C. From DWBA entrance to CDCC entrance

The inclusive sum rule was originally evaluated with a distorted-wave entrance channel. Replacing that entrance wave with the full three-body CDCC wave function, thirty-two years after the framework was first written down, produced the first inclusive-breakup calculation that is consistent at the three-body level, and identified incomplete fusion with one-step direct capture (Lei and Moro, 2019b, 2023). A new angular-momentum-basis evaluation of the sum-rule integral (Lei, 2018), a careful treatment of its convergence that contrasts the bin method with the Vincent-Fortune contour (Lei, 2025), a diagnostic that tests the surface approximation by radially cutting the scattering wave functions (Liu *et al.*, 2023a), and a head-to-head comparison of the inclusive-breakup model against a Glauber-with-quantum-scattering approach (Liu *et al.*, 2023b) have together turned the inclusive sum rule into a controlled, benchmarked tool.

D. Spectroscopy, knockout, and surrogates

Inclusive observables are also how structure is extracted. Knockout and transfer reactions probe spectroscopic factors, and the apparent quenching of those factors can be of dynamical rather than structural origin, arising from induced non-additive and polarization terms that an additive optical potential omits; the surface-integral reformulation of the transfer amplitude makes explicit how much of the result is carried by the nuclear interior versus the surface (Escher *et al.*, 2014). The Trojan Horse method inverts a quasi-free three-body transfer to recover a sub-Coulomb binary cross section relevant to astrophysics, with the extracted quantity being half-off-shell and the binary subreaction proceeding as if at sub-Coulomb energy (Tribble *et al.*, 2014; Tumino *et al.*, 2025; A. *et al.*, 2008). The interpretation of the extracted cross section requires care with the Coulomb renormalization of the half-off-shell vertex (A.M. *et al.*, 2019), and the method continues to deliver near-threshold resonance strengths inaccessible to direct measurement, as in the recent determination of a low-energy resonance in $^{19}\text{F}(p, \alpha\gamma)^{16}\text{O}$ (X.D. *et al.*, 2025). The surrogate-reaction method uses one reaction to stand in for another that cannot be measured directly (Escher *et al.*, 2012), and the broader role of these indirect three-body techniques in stellar reaction networks is surveyed in a recent review (Wiescher *et al.*, 2025). In each case the three-body amplitude is the parent object, and the inclusive observable is a particular projection of it, which is the link between this hallmark and the synthesis of Sec. IX.

VI. HALLMARK 5: UNIVERSALITY AND SCALE SEPARATION

A. The organizing idea

The fifth hallmark steps back from any particular reaction to ask what is universal. Near threshold, the low-energy behavior of a two-body subsystem is governed by its scattering length and effective range, and a large scattering length signals a shallow bound or virtual state through the pole structure of the effective-range expansion. When two-body scattering lengths are large compared with the range of the interaction, three-body observables become nearly independent of the short-range details and are controlled by a single additional three-body parameter. This is the regime of the Efimov effect (Efimov, 1970), in which an infinite geometric tower of three-body bound states accumulates at threshold, and of the broader universality reviewed comprehensively by Braaten and Hammer (Braaten and Hammer, 2006). The nuclear manifestation is the Phillips line, the model-independent correlation between the doublet nucleon-deuteron scattering length and the triton binding energy,

and the systematic organization of these correlations is the subject of the three-body-force review of Hammer, Nogga, and Schwenk (Hammer *et al.*, 2013) and the nuclear effective-field-theory review of Hammer, König, and van Kolck (Hammer *et al.*, 2020).

B. Effective field theory and the nuclear cases

Pionless and halo effective field theories make this universality systematic, organizing observables in an expansion in the ratio of the short to the long distance scale; the perturbative-pion power counting of Kaplan, Savage, and Wise is one realization for the two-nucleon sector (Kaplan *et al.*, 1998). The single most direct nuclear demonstration in the reaction context is the characterization of ^6Li as a deuteron halo (Lei *et al.*, 2018): across a wide variety of underlying interactions, the deuteron-alpha scattering length and the ^6Li separation energy collapse onto universal values, exactly as the universality picture predicts. The Borromean halos ^6He and ^{11}Li are three-body bound states with no bound two-body subsystem, the cleanest nuclear realizations of genuinely three-body binding (Zhukov *et al.*, 1993). The same logic, tracking an S-matrix pole as an interaction strength is scaled, classifies the bound, virtual, and resonant states of systems too broad to find directly, and has recently been applied to strangeness-bearing three- and four-body systems.

C. Why universality matters for reactions

Universality is not a separate subfield bolted onto reaction theory; it is the principle that lets one transfer intuition and constraint between systems. It tells the reaction theorist which features of a calculation are robust and which are artifacts of a particular potential, it supplies the low-energy input that constrains the interactions fed into the exact and approximate dynamics of the first two hallmarks, and it connects the nuclear three-body problem to the cold-atom and hypernuclear realizations where the same scaling laws hold. The link in Fig. 1 from this hallmark back to exactness is precisely this: universality is the language in which the inputs to a three-body calculation acquire meaning beyond a single fit.

VII. ENABLING CHARACTERISTIC 1: ASYMPTOTICS AND BOUNDARY CONDITIONS

The hardest part of a three-body calculation is often not the interior dynamics but the asymptotics. Three-body Coulomb has no closed-form asymptotic wave function, and the open breakup continuum makes outgoing-wave matching delicate. Every hallmark therefore rests

on a boundary-condition technology, and these technologies are worth treating as a coherent body of technique.

Complex scaling rotates the radial coordinate into the complex plane so that outgoing waves decay and a scattering problem becomes a bound-state-like problem on a square-integrable basis, with a validity condition relating the rotation angle to the channel momentum and a spectral foundation in the Balslev-Combes theorem (Balslev and Combes, 1971). Applied with care it reproduces exact momentum-space results to several digits for nucleon-triton scattering above breakup and for optical-model reactions (Lazauskas, 2019), and it has become the basis of a class of fast scattering emulators discussed in the next section (Liu *et al.*, 2024). The Lagrange-mesh method of Baye and Heenen (Baye and Heenen, 1986) expands the wave function on functions localized at Gauss-quadrature points, giving a diagonal representation of local potentials, spectral accuracy at modest mesh sizes, and, crucially for three-body breakup, no need to multipole-expand nonlocal or multidimensional potentials; it is reviewed comprehensively in (Baye, 2015). The Lagrange-mesh R-matrix is the boundary-matching engine for inhomogeneous coupled-channels equations and for the inclusive-breakup Green’s function (Lei and Descouvemont, 2020), and the direct boundary-matching method dispenses with the auxiliary Bloch operator entirely (Lei, 2026a). The Vincent-Fortune contour (Vincent and Fortune, 1973) renders the marginally convergent post-form breakup integral tractable by deforming the radial contour into the complex plane, and the contrast between it and the bin method is one of the practical themes of inclusive-breakup convergence (Lei, 2025). For resonance observables the phenomenological R-matrix supplies the channel-radius boundary conditions, and is the post-processing engine that turns resonance parameters into thermonuclear rates (deBoer *et al.*, 2021). The three rival treatments of Coulomb in the exact equations, screening, the distorted-wave basis, and the Faddeev-Merkuriev split, are the charged-particle face of the same asymptotic problem and were discussed under the first hallmark.

The unifying observation is that the choice of boundary-condition technology is often what distinguishes one practitioner’s method from another’s, even when the underlying physics is identical. A review that treats CDCC, the inclusive sum rule, and the exact equations as separate would miss that they are, at the level of asymptotics, solving variants of one problem.

VIII. ENABLING CHARACTERISTIC 2: COMPUTATION, EMULATION, AND INFERENCE

A. The cost problem and the reduced-basis answer

A single high-fidelity three-body solve is expensive, and the tasks that now define the frontier, sensitivity analysis, calibration against data, and uncertainty quantification, require many thousands to many millions of solves. The response of the past decade has been the projection-based reduced-basis emulator: compute the exact solution at a handful of training parameter points, use those snapshots to span a low-dimensional subspace, project the governing equation onto it, and solve the small projected problem at any new parameter point. Eigenvector continuation (Frame *et al.*, 2018) and the proper-orthogonal-decomposition reduced-basis method are, as the field’s own surveys make explicit, sibling instances of a single framework (Drischler *et al.*, 2023; Bonilla *et al.*, 2022), differing mainly in how the basis is constructed and in whether the target problem is a Hermitian eigenproblem or a non-Hermitian boundary-value problem. The distinction that does matter is whether the parameters enter the operator affinely, as the chiral low-energy constants do, in which case the projection is exact in structure and gradient-based calibration of the interaction is already practical (Svensson *et al.*, 2022), or non-affinely, as the optical-potential geometry does, in which case an interpolation layer is required.

B. Scattering emulators and the differentiable turn

For reactions specifically, eigenvector continuation was extended to complex optical potentials and to three-body scattering (Furnstahl *et al.*, 2020; Zhang and Furnstahl, 2022), reduced-order emulators were built for optical-model and coupled-channel problems (Odell *et al.*, 2024; Catacora-Rios *et al.*, 2026), and the complex-scaling and complex-energy formulations were turned into emulators in their own right (Liu *et al.*, 2024; Zhang, 2025). A reduced-basis emulator for the full multichannel CDCC problem achieves a speedup of more than two orders of magnitude at sub-percent accuracy on an eighteen-parameter deuteron-induced reaction (Lei, 2026b). The most consequential recent development is the differentiable turn: rebuilding the reduced operator so that automatic differentiation supplies exact likelihood gradients enables gradient-based Hamiltonian Monte Carlo sampling of the posterior, which is what makes high-dimensional calibration of an expensive reaction model feasible, converging in minutes on a single processor where derivative-free samplers would stall (Lei, 2026c). The exact Lippmann-Schwinger reduction of the few-body emulator for affine interactions (Heihoff *et al.*, 2026) shows that in the affine case no learning is needed at all.

C. Inference and its pitfalls

The point of all this machinery is inference: combining theory with data to produce posterior distributions for parameters and predictions with honest error bars. The Bayesian methodology for nuclear effective field theory, including the treatment of truncation error, was developed largely by the BUQEYE program, and the early demonstration that a precision artifact could be mistaken for a physical threshold anomaly until a proper uncertainty analysis was done is a cautionary tale from the reaction side (Navarro Pérez and Lei, 2019). The choice of sampler matters with dimension: the affine-invariant ensemble sampler (Foreman-Mackey *et al.*, 2013) is efficient at low dimension but silently fails above roughly fifty parameters (Huijser *et al.*, 2022), which is exactly why gradient-based Hamiltonian Monte Carlo (Neal, 2011), with its automatic-tuning variant (Hoffman and Gelman, 2014), has become the tool of choice for high-dimensional posteriors and why a differentiable emulator is valuable. On the structural frontier, neural-network ansatz solvers for the few-body scattering wave function carry an a-priori bound on their cross-section error that classical solvers do not natively provide, and are designed to scale toward the regime where the exact four-body equations become prohibitive (Lawrence and Yamauchi, 2026).

This enabling characteristic closes the loop of Fig. 1. The first five hallmarks produce expensive forward maps from interaction and reaction parameters to observables; this characteristic is what makes those maps invertible against data, turning three-body reaction theory from the production of single numbers into the production of calibrated predictions.

A note on tools is in order. The calculations surveyed here are carried out with a range of community and private codes, and where a specific implementation detail matters I have described what is computed rather than naming the software; the convention throughout is that the physics, not the package, is the object of the review.

IX. SYNTHESIS: ONE AMPLITUDE, MANY OBSERVABLES

The five hallmarks are presented separately for clarity, but they describe one physical process viewed through different windows. The central claim of this synthesis is that the three-body breakup amplitude, the amplitude for the projectile to dissociate in the field of the target, is the common parent, and that the elastic, breakup, fusion, and inclusive observables of the separate hallmarks are different projections of it.

The clearest expression of this unity is the relationship between the second, third, and fourth hallmarks. CDCC and the dynamic polarization potential are the *same coupled-channels equations*: solving them and reading off

the elastic S-matrix gives the polarization picture of hallmark three, while solving them and projecting onto explicit final states gives the breakup cross sections of hallmark two. The inclusive sum rule of hallmark four then takes the CDCC entrance wave function as its source and sums coherently over the unobserved fragment, so that the very same wave function that yields elastic scattering and exclusive breakup yields, under a different final-state condition, the inclusive non-elastic breakup and the incomplete fusion. Fusion suppression, breakup spectra, and inclusive yields are not independent phenomena to be modeled separately; they are bookkeeping on a single flux. The flux-conservation analyses that decompose the absorption cross section make this explicit at the level of the optical theorem.

The exact equations of the first hallmark and the universality of the fifth bracket this structure from above and below. Faddeev/AGS is the standard against which the coupled-channels approximation is validated, so the benchmark link in Fig. 1 is what licenses the whole CDCC enterprise; the agreement of CDCC with the exact solution for deuteron reactions is the warrant for trusting CDCC where no exact solution exists. Universality, at the other end, supplies the low-energy meaning of the inputs: the scattering lengths that fix the two-body subsystems, the three-body parameter that fixes the leading three-body physics, and the model-independent correlations that tell the practitioner which features of a result are robust.

This suggests a practical map of which method answers which question. For a definitive benchmark on a light system where Coulomb is manageable, the exact momentum-space or configuration-space equations are the tool. For the routine phenomenology of a weakly bound projectile on an arbitrary target, CDCC and its extensions are the workhorse, with the choice between three-body and four-body, inert and excited target, non-relativistic and relativistic, dictated by the system. For an inclusive measurement that detects one fragment, the inclusive sum rule evaluated on a CDCC entrance wave is the appropriate instrument. For fusion and its suppression, the survival-probability formalism and the polarization-potential decomposition give the mechanism. And for any of these where a calibration or an uncertainty is required, the emulator-plus-sampler machinery of the second enabling characteristic is what makes the question answerable at all.

X. OPEN CHALLENGES

The four-body frontier. The most pressing structural challenge is the consistent extension of every hallmark to genuine four-body systems. Four-body CDCC exists and works for elastic scattering of three-body projectiles, but the exact Yakubovskiy benchmark against which to

validate it is available only for the lightest systems, the inclusive sum rule has not been formulated for a three-body projectile breaking into three fragments, and the merger of the inclusive-breakup framework with four-body CDCC for halo systems remains unrealized. The dream calculation, a fully consistent four-body inclusive-breakup treatment of a Borromean halo on a heavy target, is not yet within reach.

Coulomb in the exact equations at high Z . The three coexisting treatments of Coulomb in the Faddeev/AGS equations each have a regime of validity and none is universal. A method that handles arbitrary Z without screening artifacts and without the configuration-space cost of complex scaling would remove the principal barrier to exact benchmarks for heavy-target reactions, which is precisely where the phenomenological methods are least constrained.

Exact dynamics with realistic structure. The exact equations are usually solved with simplified two-body inputs, while realistic structure is usually treated with approximate dynamics. Closing this gap, exact three-body dynamics with microscopic, antisymmetrized cluster structure and consistent chiral interactions, is a long-standing goal that the convergence of ab initio structure methods is beginning to make plausible.

Calibration with honest uncertainties. The differentiable-emulator and Bayesian machinery is new, and its application to three-body reactions has so far been demonstrated rather than deployed at scale. The open problem is to make calibrated, uncertainty-quantified three-body reaction predictions routine, including the propagation of optical-potential and interaction uncertainties into the inclusive and fusion observables that experiments measure, and the honest accounting of the model discrepancy that the three-body truncation itself introduces.

The analyzing-power and breakup anomalies. The vector analyzing-power puzzle of nucleon-deuteron scattering, the space-star and quasi-free breakup anomalies, and the persistent tensions in polarization observables remain unexplained by current three-nucleon forces. Whether their resolution lies in the interaction, in higher-body forces, or in a piece of the reaction dynamics that current treatments omit is genuinely open.

XI. CONCLUSIONS

Quantum three-body reaction theory can look, from the outside, like a catalogue of incompatible techniques.

The argument of this review is that it is instead a single subject organized by five hallmarks: the exactness that defines correctness, the discretization that makes the continuum tractable, the dynamic coupling that gives weakly bound systems their distinctive phenomenology, the inclusive treatment that connects amplitudes to measured cross sections, and the universality that ties it all to a wider few-body physics. The two enabling characteristics, the control of asymptotics and the machinery of computation and inference, are the substrate on which the hallmarks rest. The unity is not merely organizational; it is physical, because the same breakup amplitude generates the elastic, breakup, fusion, and inclusive observables that the hallmarks treat in turn.

Where is the field heading. The next decade will, I expect, be defined by three convergences. The exact and the affordable will move closer together as emulators make high-fidelity calculations cheap enough to use where only approximations were previously possible, and as differentiable, calibrated computation becomes the default rather than the exception. The structural and the dynamical will merge as ab initio cluster structure is fed into consistent three-body and four-body reaction dynamics. And the nuclear and the universal will continue to inform each other, as the scaling laws that govern halo nuclei, cold atoms, and hypernuclei are recognized as one physics. The three-body problem will remain unsolved in the sense Newton would have understood, but as a quantitative, predictive, uncertainty-quantified instrument for reading the structure and reactions of nuclei far from stability, it is rapidly coming of age.

ACKNOWLEDGMENTS

This work was supported in part by the National Natural Science Foundation of China. I thank my collaborators across the lines of work surveyed here, whose results are cited throughout, and the broader three-body reaction-theory community whose collective effort this review attempts to organize.

REFERENCES

- M. V. Zhukov, B. V. Danilin, D. V. Fedorov, J. M. Bang, I. J. Thompson, and J. S. Vaagen, Phys. Rept. **231**, 151 (1993).
- L. D. Faddeev, Sov. Phys. JETP **12**, 1014 (1961).
- E. O. Alt, P. Grassberger, and W. Sandhas, Nucl. Phys. B **2**, 167 (1967).
- N. Austern, Y. Iseri, M. Kamimura, M. Kawai, G. Rawitscher, and M. Yahiro, Phys. Rept. **154**, 125 (1987).
- W. Glöckle, H. Witała, D. Hüber, H. Kamada, and J. Golak, Phys. Rept. **274**, 107 (1996).
- I. J. Thompson, Comput. Phys. Rep. **7**, 167 (1988).
- S. Typel and G. Baur, Phys. Rev. C **50**, 2104 (1994).
- R. E. Tribble, C. A. Bertulani, M. La Cognata, A. M.

- Mukhamedzhanov, and C. Spitaleri, Rep. Prog. Phys. **77**, 106901 (2014).
- C. Drischler, J. A. Melendez, R. J. Furnstahl, A. J. Garcia, and X. Zhang, Front. Phys. **10**, 1092931 (2023).
- C. Lovelace, Phys. Rev. **135**, B1225 (1964).
- W. Glöckle, *The Quantum Mechanical Few-Body Problem* (Springer-Verlag, Berlin, 1983).
- D. J. Ernst, C. M. Shakin, and R. M. Thaler, Phys. Rev. C **8**, 46 (1973).
- A. Deltuva, A. M. Moro, E. Cravo, F. M. Nunes, and A. C. Fonseca, Phys. Rev. C **76**, 064602 (2007).
- N. J. Upadhyay, A. Deltuva, and F. M. Nunes, Phys. Rev. C **85**, 054621 (2012).
- A. M. Mukhamedzhanov, V. Eremenko, and A. I. Sattarov, Phys. Rev. C **86**, 034001 (2012).
- S. P. Merkuriev, Ann. Phys. (NY) **130**, 395 (1980).
- R. Lazauskas and J. Carbonell, Phys. Rev. C **84**, 034002 (2011).
- J. Lei, L. Hlophe, C. Elster, A. Nogga, F. M. Nunes, and D. R. Phillips, Phys. Rev. C **98**, 051001(R) (2018).
- L. Hlophe, J. Lei, C. Elster, A. Nogga, and F. M. Nunes, Phys. Rev. C **96**, 064003 (2017).
- L. Hlophe, J. Lei, C. Elster, A. Nogga, and F. M. Nunes, Phys. Rev. C **100**, 034609 (2019).
- E. Epelbaum, A. Nogga, W. Glöckle, H. Kamada, U.-G. Meißner, and H. Witała, Phys. Rev. C **66**, 064001 (2002).
- Austern, Vincent, and Farrell, Ann. Phys. **114**, 93 (1978).
- Egami, Ogata, Matsumoto, Iseri, Kamimura, and Yahiro, Phys. Rev. C **70**, 047604 (2004).
- F. Nunes, A. Mukhamedzhanov, C. Rosa, and B. Irgaziev, Nuclear Physics A **736**, 255 (2004).
- K. N, K. KW, and R. K, Nucl. Phys. A **722**, 267c (2003).
- Gómez-Ramos, Moro, Gómez-Camacho, and Thompson, Phys. Rev. C **92**, 014613 (2015).
- M. Gómez-Ramos and A. M. Moro, Phys. Rev. C **95**, 034609 (2017).
- T. Matsumoto, E. Hiyama, K. Ogata, Y. Iseri, M. Kamimura, S. Chiba, and M. Yahiro, Phys. Rev. C **70**, 061601(R) (2004).
- M. Rodríguez-Gallardo, J. M. Arias, J. Gómez-Camacho, A. M. Moro, I. J. Thompson, and J. A. Tostevin, Phys. Rev. C **77**, 064609 (2008).
- S. Watanabe, T. Matsumoto, K. Ogata, and M. Yahiro, Phys. Rev. C **92**, 044611 (2015).
- Y. Sakuragi, M. Yahiro, and M. Kamimura, Prog. Theor. Phys. Suppl. **89**, 136 (1986).
- Y. Hirabayashi, Physical Review C **44**, 1581 (1991).
- Kubono, Sugitani, Tanaka, Morita, Sakuragi, and Kamimura, Phys. Lett. B **163**, 75 (1985).
- T. Aumann and C. A. Bertulani, Prog. Part. Nucl. Phys. **112**, 103753 (2020).
- T. Druet and P. Descouvemont, European Physical Journal A **48**, 147 (2012).
- D. Y. Pang, N. K. Timofeyuk, R. C. Johnson, and J. A. Tostevin, Phys. Rev. C **87**, 064613 (2013).
- M. Yahiro, K. Ogata, T. Matsumoto, and K. Minomo, Prog. Theor. Exp. Phys. **2012**, 01A206 (2012).
- Tostevin, Nunes, and Thompson, Phys. Rev. C **63**, 024617 (2001).
- Mortimer, Thompson, and Tostevin, Phys. Rev. C **65**, 064619 (2002).
- Ogata, Hashimoto, Iseri, Kamimura, and Yahiro, Phys. Rev. C **73**, 024605 (2006).
- D. Howell, J. Tostevin, and J. Al-Khalili, J. Phys. G: Nucl. Part. Phys. **31**, S1881 (2005).
- P. Capel, D. Baye, and V. S. Melezhik, Phys. Rev. C **70**, 064605 (2004).
- A. G. Camacho, A. Diaz-Torres, P. Gomes, and J. Lubian, Physical Review C **91**, 014607 (2015).
- A. G. Camacho, A. Diaz-Torres, P. Gomes, and J. Lubian, Physical Review C **93**, 024604 (2016).
- J. R.C., A.-K. J.S., and T. J.A., Phys. Rev. Lett. **79**, 2771 (1997).
- D. Pietro, Figuera, Amorini, Angulo, Cardella, Cherubini, Davinson, Musumarra, Pellegriti, and Raabe, Phys. Rev. C **69**, 044613 (2004).
- A. Di Pietro, G. Randisi, V. Scuderi, L. Acosta, F. Amorini, M. Borge, P. Figuera, M. Fisichella, L. Fraile, and J. Gomez-Camacho, Phys. Rev. Lett. **105**, 022701 (2010).
- J. Kolata, E. Aguilera, F. Becchetti, Chen, Yu, P. DeYoung, H. Garcia-Martinez, J. Hinnefeld, J. Lupton, E. Martinez-Quiroz, and G. Peaslee, Phys. Rev. C **69**, 047601 (2004).
- Romoli, Vardaci, D. Pietro, D. Francesco, D. Rosa, Inghima, L. Commara, Martin, Pierrousakou, and Sandoli, Phys. Rev. C **69**, 064614 (2004).
- Woolliscroft, Fulton, Cowin, Dasgupta, Hinde, Morton, and Berriman, Phys. Rev. C **69**, 044612 (2004).
- Benjamim, Lepine-Szily, M. Junior, Lichtenthaler, Guimaraes, Gomes, Chamon, Hussein, Moro, and Arazi, Phys. Lett. B **647**, 30 (2007).
- Mohr, de Faria, Lichtenthaler, Pires, and Guimaraes, Phys. Rev. C **82**, 044606 (2010).
- Aguilera, Martinez-Quiroz, Lizcano, Gomez-Camacho, Kolata, Lamm, Guimaraes, Lichtenthaler, Camargo, and Becchetti, Phys. Rev. C **79**, 021601 (2009).
- Zamora, Guimaraes, Barioni, Lepine-Szily, and Lichtenthaler, Phys. Rev. C **84**, 034611 (2011).
- A. Barioni, J. Zamora, V. Guimaraes, B. Paes, J. Lubian, E. Aguilera, J. Kolata, A. Roberts, F. Becchetti, and A. Villano, Phys. Rev. C **84**, 014603 (2011).
- I. J. Thompson, M. A. Nagarajan, J. S. Lilley, and M. J. Smithson, Nucl. Phys. A **505**, 84 (1989).
- L. F. Canto, R. Donangelo, M. S. Hussein, and G. M. Marta, Phys. Rev. C **58**, 1107 (1998).
- Hussein, Phys. Rev. C **30**, 1962 (1984).
- Yadav, Sharma, Singh, Singh, Sharma, Gupta, Kumar, Singh, Prasad, and Bhowmik, Phys. Rev. C **85**, 034614 (2012).
- B. M, Phys. Rep. **129**, 145 (1985).
- Balantekin and Takigawa, Rev. Mod. Phys. **70**, 77 (1998).
- Rowley, Timmers, Leigh, Dasgupta, Hinde, Mein, Morton, and Newton, Phys. Lett. B **373**, 23 (1996).
- Gasques, Chamon, Gomes, and Lubian, Nucl. Phys. A **764**, 135 (2006).
- D. M, H. DJ, B. RD, A. RM, B. AC, C. N, G. PRS, M. CR, N. JO, S. de Toledo A, and H. K, Phys. Rev. Lett. **82**, 1395 (1999).
- D. M, H. DJ, H. K, M. SB, G. PRS, A. RM, B. RD, B. AC, C. N, M. CR, N. JO, and S. de Toledo A, Phys. Rev. C **66**, 041602 (2002).
- Dasgupta, Gomes, Hinde, Moraes, Anjos, Berriman, Butt, Carlin, Lubian, Morton, Newton, and S. de Toledo, Phys. Rev. C **70**, 024606 (2004).
- Rath, Santra, Singh, Tripathi, Parkar, Nayak, Mahata, Palit, and Choudhury, Phys. Rev. C **79**, 051601 (2009).
- Trotta, Sida, Alamanos, Andreyev, Auger, Balabanski, Borcea, Coulier, Drouart, and Gillibert, Phys. Rev. Lett. **84**, 2342 (2000).
- Diaz-Torres and Thompson, Phys. Rev. C **65**, 024606 (2002).

- Diaz-Torres, Thompson, and Beck, *Phys. Rev. C* **68**, 044607 (2003).
- H. K, V. A, D. CH, and L. SM, *Phys. Rev. C* **61**, 037602 (2000).
- K. N, K. KW, and R. K, *Phys. Rev. C* **65**, 014601 (2001).
- Gomes, Rodriguez, Marti, Padron, Chamon, F. Niello, Capurro, Pacheco, Testoni, Arazi, Lubian, and Hussein, *Phys. Rev. C* **71**, 034608 (2005).
- Beck, Keeley, and Diaz-Torres, *Phys. Rev. C* **75**, 054605 (2007).
- Beck, *Nucl. Phys. A* **787**, 251c (2007).
- Beck, Rowley, Papka, Courtin, Rousseau, Keeley, Diaz-Torres, and Hagino, *Nucl. Phys. A* **834**, 440c (2010).
- Souza, Beck, Carlin, Keeley, L. Neto, de Moura, and Munhoz, *Nucl. Phys. A* **821**, 36 (2009).
- Lubian, Correa, Aguilera, Canto, Gomez-Camacho, Quiroz, and Gomes, *Phys. Rev. C* **79**, 064605 (2009).
- Canto, Gomes, Lubian, Chamon, and Crema, *J. Phys. G: Nucl. Part. Phys.* **36**, 015109 (2009a).
- Aguilera, Kolata, Nunes, Becchetti, and DeYoung, *Phys. Rev. Lett.* **84**, 5058 (2000).
- Satchler, Nagarajan, Lilley, and Thompson, *Ann. Phys.* **178**, 110 (1987).
- Satchler, *Nucl. Phys. A* **472**, 591 (1987).
- G. R. Satchler, *Phys. Rept.* **199**, 147 (1991).
- Keeley, Christley, Clarke, Fulton, and Thompson, *Nucl. Phys. A* **582**, 314 (1995).
- Hussein, Gomes, Lubian, and Chamon, *Phys. Rev. C* **73**, 044610 (2006).
- P. Gomes, R. Anjos, C. Muri, J. Lubian, I. Padron, L. Chamon, R. Liguori Neto, and N. Added, *Phys. Rev. C* **70**, 054605 (2004).
- Figueira, Abriola, Fernandez-Niello, Arazi, Capurro, de Barbara, Marti, Martinez-Heimann, Pacheco, Testoni, Padron, Gomes, and Lubian, *Phys. Rev. C* **73**, 054603 (2006).
- Figueira, Fernandez-Niello, Abriola, Arazi, Capurro, de Barbara, Marti, Martinez-Heimann, Negri, Pacheco, Padron, Gomes, Lubian, Correa, and Paes, *Phys. Rev. C* **75**, 017602 (2007).
- Kumawat, Jha, Roy, Parkar, Santra, Kumar, Dutta, Shukla, Pant, Mohanty, Choudhury, and Kailas, *Phys. Rev. C* **78**, 044617 (2008).
- G. Camacho, Aguilera, M. Quiroz, Gomes, Lubian, and Canto, *Nucl. Phys. A* **833**, 156 (2010).
- Deshmukh, Mukherjee, Nayak, Biswas, Santra, Mirgule, Appannababu, Patel, Saxena, Choudhury, Lubian, and Gomes, *Eur. Phys. J. A* **47**, 118 (2011a).
- Deshmukh, Mukherjee, Patel, Singh, Rath, Nayak, Biswas, Santra, Mirgule, Danu, Gupta, Saxena, Choudhury, Kumar, Lubian, Lopes, Cardozo, and Gomes, *Phys. Rev. C* **83**, 024607 (2011b).
- S. A, K. S, S. P, C. A, N. A, S. AM, R. R. V, M. S, D. SK, and A. DK, *Nucl. Phys. A* **635**, 411 (1998).
- M. Zadro, P. Figuera, A. Di Pietro, F. Amorini, M. Fisichella, O. Goryunov, M. Lattuada, C. Maiolino, A. Musumarra, and V. Ostashko, *Phys. Rev. C* **80**, 064610 (2009).
- J. Figueira, J. Fernandez Niello, A. Arazi, O. Capurro, P. Carnelli, L. Fimiani, G. Marti, D. Martinez Heimann, A. Negri, A. Pacheco, J. Lubian, D. Monteiro, and P. Gomes, *Phys. Rev. C* **81**, 024613 (2010).
- Signorini, Andrighetto, Guo, Ruan, Stroe, Soramel, and Thompson, *Nucl. Phys. A* **701**, 23c (2002).
- Biswas, Roy, Sinha, and Mukherjee, *Nucl. Phys. A* **802**, 67 (2008).
- Parkar, Jha, Roy, Santra, Ramachandran, Shrivastava, Chatterjee, Jain, Jain, and Kailas, *Phys. Rev. C* **78**, 021601 (2008).
- Keeley, Kemper, and Rusek, *Phys. Rev. C* **66**, 044605 (2002).
- Rusek, Keeley, Kemper, and Raabe, *Phys. Rev. C* **67**, 041604 (2003).
- Keeley, Cook, Kemper, Roeder, Weintraub, Marechal, and Rusek, *Phys. Rev. C* **68**, 054601 (2003).
- Pakou, Alamanos, Gillibert, Kokkoris, and Kossionides, *Phys. Rev. Lett.* **90**, 202701 (2003).
- Pakou and Rusek, *Phys. Rev. C* **69**, 057602 (2004).
- L. F. Canto, J. Lubian, P. R. S. Gomes, and M. S. Hussein, *Phys. Rev. C* **80**, 047601 (2009b).
- A. Gómez Camacho, B. Wang, and H. Q. Zhang, *Phys. Rev. C* **97**, 054610 (2018).
- L. F. Canto, P. R. S. Gomes, R. Donangelo, and M. S. Hussein, *Phys. Rept.* **424**, 1 (2006).
- L. F. Canto, P. R. S. Gomes, R. Donangelo, J. Lubian, and M. S. Hussein, *Phys. Rept.* **596**, 1 (2015).
- N. Keeley, R. Raabe, N. Alamanos, and J. L. Sida, *Prog. Part. Nucl. Phys.* **63**, 396 (2009).
- J. Lei and A. M. Moro, *Phys. Rev. Lett.* **122**, 042503 (2019a).
- J. Lei and A. M. Moro, *Phys. Rev. C* **95**, 044605 (2017).
- H. Liu, J. Lei, and Z. Z. Ren, *Phys. Rev. C* **113**, 054601 (2026a).
- H. Liu, J. Lei, and Z. Z. Ren, *Phys. Lett. B* **877**, 140479 (2026b).
- Luong, Dasgupta, Hinde, du Rietz, Rafiei, Lin, Evers, and Diaz-Torres, *Phys. Lett. B* **695**, 105 (2011).
- So, Lee, and Kim, *Phys. Rev. C* **83**, 067602 (2011).
- Baur, Shyam, Rösels, and Trautmann, *Phys. Rev. C* **21**, 2668 (1980).
- Motobayashi, Iwasa, Ando, Kurokawa, and Murakami, *Phys. Rev. Lett.* **73**, 2680 (1994).
- Schumann, Hammache, Typel, Uhlig, and Summerer, *Phys. Rev. Lett.* **90**, 232501 (2003).
- Schümann, Typel, Hammache, Sümmerer, Uhlig, Böttcher, Cortina, Förster, Gai, and Geissel, *Phys. Rev. C* **73**, 015806 (2006).
- Esbensen and Bertsch, *Nucl. Phys. A* **600**, 37 (1996).
- Kikuchi, Motobayashi, Iwasa, Ando, and Kurokawa, *Phys. Lett. B* **391**, 261 (1997).
- B. Davids, S. Austin, D. Bazin, H. Esbensen, B. Sherrill, I. Thompson, and J. Tostevin, *Phys. Rev. C* **63**, 065806 (2001).
- Esbensen and Bertsch, *Phys. Rev. C* **59**, 3240 (1999).
- Dasso, Lenzi, and Vitturi, *Nucl. Phys. A* **639**, 635 (1998).
- Nunes and Thompson, *Phys. Rev. C* **59**, 2652 (1999).
- Esbensen and Hencken, *Phys. Rev. C* **61**, 054606 (2000).
- Kolata, Guimaraes, Peterson, Santi, White-Stevens, Vincent, Becchetti, Lee, O'Donnell, Roberts, and Zimmerman, *Phys. Rev. C* **63**, 024616 (2001).
- Descouvemont, *Phys. Rev. C* **70**, 065802 (2004).
- Lehr, Aumann, Duer, Saito, Nakamura, and et al., *Phys. Rev. Lett.* **136**, 172501 (2026).
- N. Austern and C. M. Vincent, *Phys. Rev. C* **23**, 1847 (1981).
- M. Ichimura, N. Austern, and C. M. Vincent, *Phys. Rev. C* **32**, 431 (1985).
- A. K. Kerman and K. W. McVoy, *Ann. Phys. (NY)* **122**, 197 (1979).
- Baur, Shyam, Rösels, and Trautmann, *Phys. Rev. C* **28**, 946 (1983).
- Shyam, Baur, Rösels, and Trautmann, *Phys. Rev. C* **22**, 1401 (1980).

- Udagawa, Li, and Tamura, Phys. Lett. B **135**, 333 (1984).
 Kasano and Ichimura, Phys. Lett. B **115**, 81 (1982).
 T. Udagawa and T. Tamura, Phys. Rev. C **24**, 1348 (1981).
 X.-H. Li, T. Udagawa, and T. Tamura, Phys. Rev. C **30**, 1895 (1984).
 J. Lei and A. M. Moro, Phys. Rev. C **92**, 061602(R) (2015a).
 J. Lei and A. M. Moro, Phys. Rev. C **97**, 011601(R) (2018).
 J. Lei and A. M. Moro, Phys. Rev. C **92**, 044616 (2015b).
 Cotanch, Nucl. Phys. A **842**, 48 (2010).
 Potel, Nunes, and Thompson, Phys. Rev. C **92**, 034611 (2015).
 J. Lei and A. M. Moro, Phys. Rev. Lett. **123**, 232501 (2019b).
 J. Lei and A. M. Moro, Phys. Rev. C **108**, 034612 (2023).
 J. Lei, Phys. Rev. C **97**, 034628 (2018).
 J. Lei, Phys. Rev. C **112**, 014609 (2025).
 J. Liu, J. Lei, and Z. Z. Ren, Phys. Rev. C **108**, 024606 (2023a).
 H. Liu, J. Lei, Z. Z. Ren, *et al.*, Phys. Rev. C **108**, 014617 (2023b).
 J. E. Escher, I. J. Thompson, G. Arbanas, C. Elster, V. Ere-
 menko, L. Hlophe, and F. M. Nunes, Physical Review C **89**, 054605 (2014).
 A. Tumino, C. Spitaleri, M. La Cognata, R. G. Pizzone,
 L. Lamia, *et al.*, Prog. Part. Nucl. Phys. **143**, 104164
 (2025).
 T. A., S. C., M. A., R. G.G., C. L., C. S., C. V., E. Z., F. Z.,
 G. L., G. M., G. Gy., K. G., L. C. M., L. L., O. A., P. R.G.,
 R. S., S. M.L., and S. E., Phys. Rev. C **78**, 064001 (2008).
 M. A.M., P. D.Y., and K. A.S., Phys. Rev. C **99**, 064618
 (2019).
 S. X.D., L. C. M., V. N., R. G.G., M. M., O. A.A., R. L.,
 Z. G.L., B. B., C. S., C. M., D. P. A., F. P., G. M., G. G.L.,
 H. S., I. I., L. C. M., L. L., L. D., P. S., P. R.G., P. S.M.R.,
 R. S., S. M.L., S. R., S. C., T. D., T. O., and T. A., Phys.
 Rev. Lett. **135**, 182701 (2025).
 J. E. Escher, J. T. Burke, F. S. Dietrich, N. D. Scielzo,
 I. J. Thompson, and W. Younes, Rev. Mod. Phys. **84**, 353
 (2012).
 Wiescher, Bertulani, Brune, deBoer, Diaz-Torres, Gasques,
 Langanke, Navratil, Nazarewicz, Okolowicz, Phillips,
 Ploszajczak, Quaglioni, and Tumino, Rev. Mod. Phys. **97**,
 025003 (2025).
 V. Efimov, Phys. Lett. B **33**, 563 (1970).
 E. Braaten and H.-W. Hammer, Phys. Rept. **428**, 259 (2006).
 H.-W. Hammer, A. Nogga, and A. Schwenk, Rev. Mod. Phys.
85, 197 (2013).
 H.-W. Hammer, S. König, and U. van Kolck, Rev. Mod. Phys.
92, 025004 (2020).
 D. B. Kaplan, M. J. Savage, and M. B. Wise, Phys. Lett. B
424, 390 (1998).
 E. Balslev and J. M. Combes, Commun. Math. Phys. **22**, 280
 (1971).
 R. Lazauskas, Habilitation thesis, Univ. de Strasbourg;
 arXiv:1904.04675 (2019).
 J. Liu, J. Lei, and Z. Z. Ren, Phys. Lett. B **858**, 139070 (2024).
 D. Baye and P.-H. Heenen, J. Phys. A **19**, 2041 (1986).
 D. Baye, Phys. Rept. **565**, 1 (2015).
 J. Lei and P. Descouvemont, Phys. Rev. C **102**, 014608
 (2020).
 J. Lei, Phys. Rev. C **113**, 024614 (2026a).
 C. M. Vincent and H. T. Fortune, Phys. Rev. C **8**, 1084
 (1973).
 R. J. deBoer *et al.*, Phys. Rev. C **103**, 055815 (2021).
 D. Frame, R. He, I. Ipsen, D. Lee, D. Lee, and E. Rrapaj,
 Phys. Rev. Lett. **121**, 032501 (2018).
 E. Bonilla, P. Giuliani, K. Godbey, and D. Lee, Phys. Rev. C
106, 054322 (2022).
 I. Svensson, A. Ekström, and C. Forssén, Phys. Rev. C **105**,
 014004 (2022).
 R. J. Furnstahl, A. J. Garcia, P. J. Millican, and X. Zhang,
 Phys. Lett. B **809**, 135719 (2020).
 X. Zhang and R. J. Furnstahl, Phys. Rev. C **105**, 064004
 (2022).
 D. Odell, P. Giuliani, K. Beyer, M. Catacora-Rios, M. Y.-H.
 Chan, E. Bonilla, R. J. Furnstahl, K. Godbey, and F. M.
 Nunes, Phys. Rev. C **109**, 044612 (2024).
 M. Catacora-Rios *et al.*, Phys. Rev. C **113**, 044623 (2026).
 X. Zhang, Phys. Rev. Lett. **135**, 242501 (2025).
 J. Lei, Phys. Rev. C **113**, 044610 (2026b).
 J. Lei, arXiv:2605.30980 (submitted to Phys. Rev. Lett.)
 (2026c).
 P. Heihoff, A. A. Filin, and E. Epelbaum, arXiv:2604.25792
 (2026).
 R. Navarro Pérez and J. Lei, Phys. Lett. B **795**, 200 (2019).
 D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman,
 Publ. Astron. Soc. Pac. **125**, 306 (2013).
 D. Huijser, J. Goodman, and B. J. Brewer, Aust. N. Z. J.
 Stat. **64**, 1 (2022).
 R. M. Neal, Handbook of Markov Chain Monte Carlo, Chap-
 man & Hall/CRC, 113 (2011).
 M. D. Hoffman and A. Gelman, J. Mach. Learn. Res. **15**, 1593
 (2014).
 T. Lawrence and Y. Yamauchi, arXiv:2605.27807 (2026).